Christoph B. Kassir

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Education

University of Toronto Sep 2023–Apr 2028
B.A.Sc. Electrical and Electronics Engineering *Toronto, ON*

- UofT IEEE × Cognichip IC Hack **1st Place**, ECE295 × NWS Design Competition **1st Place**
- NSERC Research Award, First-Year Research Award, Dept & Faculty Entrance Scholarships, Dean's List
- Courses: Analog Data Converter ICs (**16nm FinFET**), VLSI Layout (**65nm CMOS**), Digital ICs (250nm CMOS), Fields & Waves, Power Electronics, Semiconductor Devices, Analog ICs, Digital Design, Computer Organization

Experience

Analog Mixed-Signal Layout Design Engineer (Intern) May 2026–Present
Marvell Technology (Celestial AI acquired) *Toronto, ON*

- **Owning design** of two major high-speed layouts (Virtuoso) for **Photonic Fabric™ SERDES RX** in **3nm finFET**.
- Architecting high-signal-density and matching-sensitive blocks. Using Marvell digital layout and extraction software.
- Designed a novel highly-parameterizable powergrid structure soon to be **used company-wide**.
- Built and presented agentic LLMs for automated generation of SKILL code for powergrid PCELLs and algorithms.
- Technology node pathfinding: device characterization and S-parameter simulations (Spectre).
- Hosting design reviews and collaborating with designers and layouters across timezones. Exposure to **2nm GAAFET/NSFET**, and proprietary silicon photonic and analog design architectures. Allocating contractor resources.

Power Electronics Engineer (Intern) Jan–Aug 2025
Myant Medtech *Toronto, ON*

- Full RTL design + verification for MHz-range FPGA converter controller (SystemVerilog, Quartus).
- Design + bring-up, neurostimulator mixed-signal PCB (Altium): digital delay circuits, DC-DC, OV/OC protection.
- Design + bring-up, ramper-reducer mixed-signal PCB: impedance matching, USB-C PD, DC-DC, OV/OC protection.
- Planar/2D transformer PCB design and magnetic simulation (PLECS).
- Overhauled 2016 legacy firmware, rewrote BLE codebase in Zephyr-RTOS (C, Linux).

ML Research Engineer (Intern) May–Aug 2025
Department of National Defense (DND) *Toronto, ON*

- Built Hidden-Markov-tuned transformers (PyTorch) on an explainable-AI VIEWS25 contract.

Projects

HBT Ultra-Low Noise Amplifier RFIC for Quantum Read-Out (Tapeout Greenlit)

- ULNA in **IHP130nm** SG13G2 **SiGe HBT** operating in deep cryostat and subthreshold, RFIC targetting 100+ GHz.
- **Winner**: UofT IEEE IC Competition. Selected for a fully-funded GlobalFoundries tapeout.

Neurocore: Mixed-Signal IC (Sim) + ASIC (Tapeout Arriving)

- **Winner, 1st place**: UofT IEEE × Cognichip IC Hack. Won digital segment TinyTapeout shuttle.
- Analog IC (Virtuoso): Custom ADC architecture has regenerative SRAM bits to withstand strong EM noise. Full asynchronous datapath and SRAM control timing verification (Spectre). **GF65nm**.
- Digital IC (SystemVerilog): RTL-to-GDSII, time-shared DWT, LSK interface, power/clock gating. **SKY130nm**.

CMOS EKV-based Simulator (Ongoing Project)

- CMOS simulator in C++17, emphasis on **moderate inversion** simulation using the EKV equations.

FPGA FFT: RTL vs HLS Comparison

- 8-point FFT unit for HLS vs RTL comparison (Vivado, HLS, SystemVerilog) + testbenches.
- Containerized solution (Bash, Docker) for Vivado installation on Apple Silicon.

PCB Wireline Audio Communication

- **Winner, 1st Place**: ECE295 × NWS Design Competition.
- Bare-metal (C) for 4kHz A-D-A conversion on-PCB (Altium). PyVisa test-automated.